

Automated compliance reporting for automotive digital product passports using interoperable data ecosystems aligned with Gaia-X and EU ESPR requirements.

Manufacturing × Product traceability and digital product passport × Cloud data platform · OS018 v3 · refreshed 2026-08-18

ATTRACTIVENESS	RIGHT TO WIN	COMPETITION	SERVICEABLE MARKET	HORIZON	PORTFOLIO
68 is the world moving	61 can we play, can we win	HIGH 0 named in the evidence	€2.0bn observed evidence · per year	NEXT regulation adopted or propose...	L2 20 signals

Attractiveness and right to win are scored and displayed separately and are never combined (SC-12). Competition is a fourth quantity, not a discount on either.

The opportunity

This opportunity helps automotive manufacturers automate compliance reporting for the EU Digital Product Passport (DPP) using interoperable data ecosystems aligned with Gaia-X. It exists now because the EU ESPR mandates DPPs for electronics and appliances by 2027, and the EU has launched the DPP Registry, creating urgent data demands. For a European automotive OEM, this means turning fragmented supply chain data into a compliant, auditable DPP without manual effort.

Market opportunity

Method	Total (TAM)	Serviceable (SAM)	Obtainable (SOM)	Confidence
Bottom-up: enterprises x adoption x contract value	€2.7bn €490m – €14.6bn	€2.0bn €352m – €10.5bn	€34.2m €6.2m – €184m	observed
Observed: tendered contract value, annualised	€20.9m €20.9m – €56.6m	€20.9m €20.9m – €56.6m	€367k €367k – €991k	observed

All figures are annual, in EUR. Ranges compound the uncertainty of each factor below; the base case is the middle column of each.

Bottom-up: enterprises x adoption x contract value

Factor	Value	Basis	Source	As of
Enterprises in Manufacturing (EU27_2020)	238,122 enterprises	observed	Eurostat · sbs_sc_ovw	2024
Enterprises adopting Cloud data platform	20.6 % of enterprises (10+ employees)	observed	Eurostat · isoc_cicce_usen2 · E_CC_PDB	2025
Annual value of one engagement	€337k per enterprise per year	observed	TED (Tenders Electronic Daily) · ted	2026-05-19..2026-08-05
Size-weighted buyer base	39,220 enterprise-equivalents at large-enterprise engagement value	assumption	config/sizing.yaml · size_class_value_weights	size-2026-08-a
Assumed obtainable share of the serviceable market	1.8 % of SAM	assumption	config/sizing.yaml · obtainable_share	size-2026-08-a

Read this before quoting the number. Contract values are observed from EU public procurement (TED). For private-sector verticals they are used as a proxy for comparable enterprise deal size: the buying entity differs, the scope of work is comparable. Engagement value is scaled by enterprise size class (10-19: x0.04, 20-49: x0.09, 50-249: x0.35, GE250: x1.0), because the observed contract value comes from large-organisation tenders.

Observed: tendered contract value, annualised

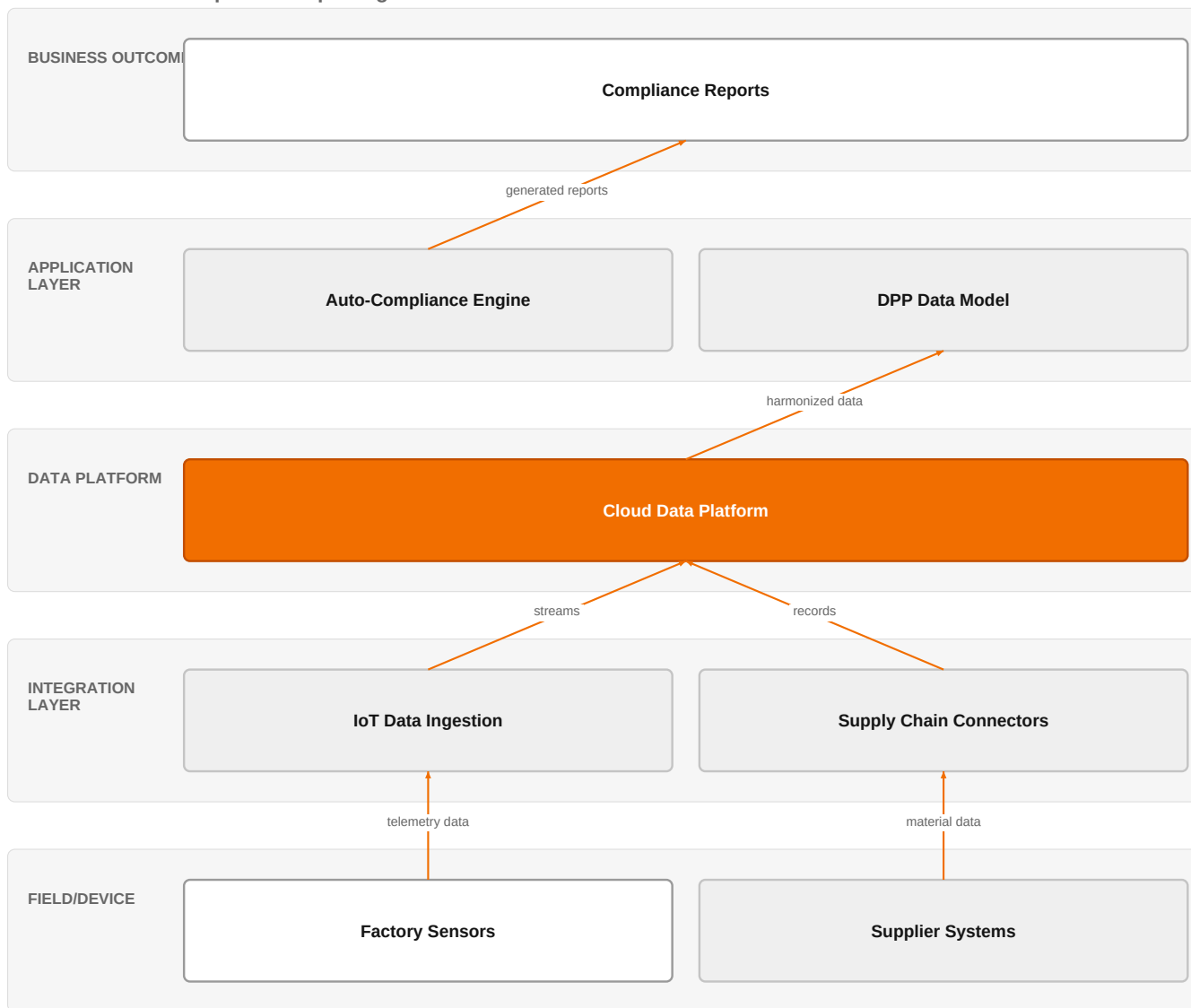
Factor	Value	Basis	Source	As of
Tendered contract value in matching CPV categories	€20.9m per year	observed	TED (Tenders Electronic Daily) · ted	2026-05-19..2026-08-05
Assumed obtainable share of the serviceable market	1.8 % of SAM	assumption	config/sizing.yaml · obtainable_share	size-2026-08-a

Read this before quoting the number. Observed over 85 days of TED coverage and scaled to a year (x4.3); a short collection window makes that scaling rough. 14 matching notices, matched on vertical via the CPV crosswalk — §4.5.2's warning applies: a crosswalk error lands directly in this number. Public procurement only. Private-sector demand for the same capability is not visible here at all, so this is a floor rather than a

market size.

The solution, in one picture

Automated DPP Compliance Reporting



■ Orange asset ■ Partner ■ Customer-owned ■ Third party / to be sourced

This diagram shows how Orange's cloud data platform and expertise turn factory and supplier data into automated DPP compliance reports.

Boxes marked as Orange assets are validated against the linked business graph; a box the model claimed for Orange without a matching asset is shown as third party.

What has changed

The EU's Ecodesign for Sustainable Products Regulation (ESPR, EU 2024/1781) mandates Digital Product Passports for electronics and appliances by 2027, creating an urgent need for machine-readable product lifecycle data. The EU has launched the Digital Product Passport Registry, raising new data demands and clarifying its operation. Gaia-X architectural frameworks are being leveraged to ensure transparency, openness, reliability, security, and interoperability in data exchange with auto-compliance tools for DPPs. Research highlights the need for semantic interoperability and integration of DPPs with digital twins and IoT to overcome technical challenges.

Sources: SIG-BA196FF05244 Springer Science and B 2026-02-03; SIG-18A289074F15 Open Access Government 2026-07-24; SIG-A3D483674C44 Packaging Insights 2026-08-03; SIG-A3AAC433E98C HKTDC Research 2026-05-07; SIG-63476A345BB2 cordis.europa.eu 2026-06-01; SIG-F47C31AB60CB MDPI AG 2026-02-14

Who buys, and why now

The Chief Compliance Officer or Head of Sustainability at a European automotive OEM signs, driven by the regulatory deadline of 2027. The pain is felt by the compliance and supply chain teams who currently manage fragmented data from tier-1 and tier-2 suppliers, making it impossible to produce a complete, accurate DPP without manual reconciliation. The

trigger is the EU DPP Registry going live and the first audit or customer request for a DPP that exposes gaps in data availability and quality.

Why it is hot now — every claim bound to a dated source

- Gaia-X architectural frameworks and proven practices are used to ensure transparency and interoperability in data exchange with auto-compliance tools for DPPs. [SIG-63476A345BB2]
- COMPLIANCE4DPP will leverage Gaia-X architectural frameworks to ensure transparency, openness, reliability, security, and interoperability in data exchange with auto-compliance tools along the value chain with digital product passport. [SIG-63476A345BB2]
- The digital product passport is examined as an integrative tool embedding sustainability, open innovation, and knowledge management into organizational practice in the fashion and cosmetics industries. [SIG-6B43CC88BC8E]
- The urgency of addressing the growing e-waste crisis and the regulatory push for circular solutions demand innovative tools like Digital Product Passports. [SIG-DFE69DA138F9]
- Plastics are a central focus of the EU Commission's Circular Economy Action Plan, and Digital Product Passports enable transparency and data sharing across a product's life cycle. [SIG-CC28A63D0BB7]
- Consumer preferences for labeling using Digital Product Passports are being analyzed, indicating potential as eco-labels. [SIG-FC25886E23D0]
- The implementation of DPPs in the cosmetics industry is explored, a sector marked by complex formulations and global supply chains. [SIG-8123897AE6E6]
- Gaia-X architectural frameworks are leveraged to ensure transparency, openness, reliability, security, and interoperability in data exchange with auto-compliance tools for DPP. [SIG-63476A345BB2]

What Orange would deliver, and why it can win

What Orange would deliver

Orange would assemble a solution using its Cloud data platform as the core, integrating IoT experts to capture real-time product lifecycle data from the factory floor and supply chain. Digital experts would design the DPP data model and workflows, while AI/Data experts would build the auto-compliance reporting engine. The engagement would start with a data readiness assessment, then deploy the cloud platform to aggregate and harmonize data, and finally configure the DPP generation and reporting aligned with Gaia-X interoperability principles.

Why Orange can win

Orange can win because it combines a certified cloud data platform (ISO 27001, TISAX) with deep integration expertise from its AI/Data and IoT experts, and it can leverage its partner ecosystem (Microsoft, AWS, Google Cloud) to provide a multi-cloud approach. The TISAX certification is particularly relevant for automotive clients, and Orange's experience with large industrial clients (e.g., Saint-Gobain Glass) demonstrates credibility. However, the assets are not unique; the differentiator is the ability to orchestrate a compliant, interoperable data ecosystem that hyperscalers alone may not provide.

Named assets behind this — each individually inspectable (LK-08)

Asset	Type	Link	Why it is linked	Curator
Microsoft	partner	L2	partner provides the required technology	unconfirmed
AWS	partner	L2	partner provides the required technology	unconfirmed
Google Cloud	partner	L2	partner provides the required technology	unconfirmed
Cloud data platform	technology	L3	no offer path, but adjacent assets or Orange research exist	unconfirmed
ISO 27001	certification	SUP	certification Orange holds is required by this vertical	unconfirmed
TISAX	certification	SUP	certification Orange holds is required by this vertical	unconfirmed
AI / Data / Cloud experts	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
IoT experts	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed

Digital experts	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Healthcare experts	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Orange Group researchers	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Colgate-Palmolive	reference	SUP	published reference in this vertical, different use case	unconfirmed
Mondelez International	reference	SUP	published reference in this vertical, different use case	unconfirmed
Saint-Gobain Glass	reference	SUP	published reference in this vertical for this use case	unconfirmed
Skytale	reference	SUP	published reference in this vertical, different use case	unconfirmed

6 further published reference(s) in this vertical are linked to this topic and are listed in the radar.

Competitive landscape

HIGH competitive intensity (score 8.6; 0 of 8 listed players appear in this topic's own evidence)

Crowded, with heavyweight incumbents already visible in the evidence. Win on a specific differentiator or do not bid.

Competitor	Category	Position here	Basis	Evidence
AWS	Hyperscaler	sells Cloud data platform; competes in Cloud, OX: Smart Industries — also an Orange partner	structural	—
Google Cloud	Hyperscaler	sells Cloud data platform; competes in Cloud — also an Orange partner	structural	—
Microsoft	Hyperscaler	sells Cloud data platform; competes in Cloud — also an Orange partner	structural	—
Accenture	Global systems integrator	sells Cloud data platform; active in Manufacturing; competes in Cloud, OX: Smart Industries	structural	—
Capgemini	Global systems integrator	sells Cloud data platform; active in Manufacturing; competes in Cloud, OX: Smart Industries	structural	—
DXC Technology	Global systems integrator	sells Cloud data platform; competes in Cloud	structural	—
Infosys	Global systems integrator	sells Cloud data platform; competes in Cloud	structural	—
Sopra Steria	Global systems integrator	sells Cloud data platform; competes in Cloud	structural	—

evidenced means this topic's own sources name the competitor — the signal ids are in the last column and are dated and clickable in the radar. **structural** means the curated register says they sell this technology into this vertical, which is true but is not proof they are in this deal. Register version competitors-2026-08-a.

Qualifying questions for the first meeting

- 1 How are you currently preparing for the EU DPP mandate for your products, and what is your target compliance date?
- 2 Which of your product lines will be in scope first, and what is the current state of your supply chain data for those products?
- 3 Do you have a data governance framework in place that can support sharing DPP data with external stakeholders?
- 4 Have you already selected a cloud provider or platform for your DPP initiative, and if so, which one?
- 5 What is your biggest pain point today: data collection, data integration, or reporting/audit preparation?
- 6 Are you considering using Gaia-X or other interoperability standards for your DPP data exchange?

Objections you will meet

They will say	You can answer
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We already have a cloud platform from AWS/Microsoft/Google; why do we need Orange?	That's a fair point. We can work with your existing cloud provider—we have partnerships with all major hyperscalers. What we add is the industry-specific expertise in DPP compliance, IoT integration, and data interoperability that hyperscalers don't provide out of the box.
The DPP requirements are still unclear; we're waiting for more regulation.	You're right that some details are still being clarified, but the registry is live and the 2027 deadline is approaching. Starting now with a flexible, interoperable architecture will save you from a costly rush later. Our approach is designed to adapt as regulations evolve.
We have our own IT team that can build this.	Building a compliant DPP system is complex, involving semantic data integration, IoT connectivity, and regulatory reporting. Our team has done this for other industries and can accelerate your time-to-compliance while your IT team focuses on core business.

Risks and unknowns

The main risk is that the regulatory landscape is still evolving; the DPP registry has just launched and detailed rules are being clarified, which could change requirements. There is also uncertainty about the level of automation achievable given the fragmented data quality in automotive supply chains. Additionally, the competitive intensity is high, and Orange's differentiation may be insufficient if the customer already has a preferred hyperscaler or integrator.

Next action, by role (FR-17)

Role	Do this next
Presales / Proposal	Prepare a bid brief for a European automotive manufacturer, leveraging Orange's Cloud data platform, ISO 27001 and TISAX certifications, and Microsoft/AWS/Google Cloud partnerships to propose an interoperable DPP compliance solution, referencing Saint-Gobain Glass as a proof point.
Sales	In a conversation with a tier-1 automotive supplier's head of sustainability, say: 'Orange's cloud data platform, combined with our AI and data expertise, can help you automate compliance reporting for digital product passports, ensuring alignment with Gaia-X and EU EPR requirements.'
Strategist / Innovator	Deep-dive into Gaia-X and EU EPR compliance reporting for DPPs, focusing on how Orange's Cloud data platform and AI/Data experts can enable interoperable data ecosystems, and identify a pilot opportunity with a manufacturing client in the automotive sector.

Evidence

Every claim in this brief resolves to one of these. Tier 1 is an authoritative primary source; tier 4 is vendor-published material, discounted in scoring (§4.3.7).

Signal	Date	Publisher	T	Title and link
SIG-63476A345BB2	2026-06-01	cordis.europa.eu	1	Fostering Resilient and Auto-Compliant Manufacturing Value Chains through Advanced Digital Prod...
SIG-6B43CC88BC8E	2026-03-14	openalex.org	1	Integrating knowledge management and digital product passports to foster sustainable and collab...
SIG-F47C31AB60CB	2026-02-14	MDPI AG	1	Dynamic Traceability and Reliability Management: Integrating the Digital Product Passport with ...
SIG-BA196FF05244	2026-02-03	Springer Science and Busi...	1	Digital Product Passport for Circular Economy: A Review of Semantic Web Vocabularies and Requir...
SIG-7B174A2FD055	2026-01-29	MDPI AG	1	Verisav Vocabularies: RDF/OWL Ontologies for Digital Product Passport and After-Sales Service A...
SIG-4FE21FD2ED06	2026-01-01	openalex.org	1	On the Utilization of Digital Product Passports for Intelligent Metal Sorting and Recycling
SIG-9F888B9F6305	2026-01-01	openalex.org	1	Semantic Data Integration for Digital Product Passports
SIG-CC28A63D0BB7	2026-01-01	openalex.org	1	From Compliance to Circularity: Stakeholder Perspectives on Adopting Digital Product Passports ...
SIG-DFE69DA138F9	2026-01-01	openalex.org	1	Shaping a circular future for electronics: The role of Digital Product Passports
SIG-FC25886E23D0	2025-12-29	openalex.org	1	Can Digital Product Passports become the next-generation eco-label? Analyzing consumer preferen...
SIG-8123897AE6E6	2025-12-15	openalex.org	1	Bridging digital tools and human skills: A sustainable competence framework for implementing di...
SIG-40DC35D6534F	2025-10-14	openalex.org	1	Predictive Generation of Digital Product Passport as Decision Support in Circular Networks
SIG-A2CC51B21A97	2025-06-01	cordis.europa.eu	1	Implementation of Advanced Digital Solutions to increase the circularity of PV panels throughou...
SIG-A9A7A4BCDCC6	2025-05-01	cordis.europa.eu	1	Demonstrating circular value chains for complex plastics via collection, reuse, repair, remanuf...
SIG-A3D483674C44	2026-08-03	Packaging Insights	2	EU's Digital Product Passport Registry raises new data demands - Packaging Insights
SIG-18A289074F15	2026-07-24	Open Access Government	2	EU launches Digital Product Passport registry before its 2027 rollout - Open Access Government
SIG-C4998B3557D1	2026-07-22	INSIGHT EU MONITORING	2	EU Commission launches Digital Product Passport Registry - INSIGHT EU MONITORING
SIG-370D30A66C8C	2026-07-21	EU NEIGHBOURS east	2	EU's Digital Product Passport Registry goes live - EU NEIGHBOURS east
SIG-A2579191BD7C	2026-07-01	kpmg.com	2	KPMG European Digital Product Passport Readiness Survey - kpmg.com
SIG-A3AAC433E98C	2026-05-07	HKTDC Research	2	Proposed EU Rules Clarify Operation of Digital Product Passport Registry - HKTDC Research

How this brief was made

Computed, not written. Attractiveness, right to win, competitive intensity and every figure in the market-opportunity section are computed from stored inputs and can be re-derived (DR-05, NFR-01). No language model produced a number anywhere in this document (§4.4.4).

Written, then checked. The narrative sections and the solution diagram were generated from this topic's evidence and from the named asset and competitor lists. Factual sections must cite signal ids attached to this topic; sections that failed that check, or that contained a generated quantity or an unsupplied organisation, were removed rather than rewritten.

Removed by those checks in this brief: competitive_landscape (no claim survived evidence binding); diagram (3 box(es) claimed to be an Orange asset that was not supplied — shown as third-party instead)

What	Version	When
Scores — weight set	w-2026-08-a	2026-08-18T19:08:26+00:00
Market sizing — assumptions	size-2026-08-a	2026-08-18T21:06:15+00:00
Competitor register	competitors-2026-08-a	2026-08-18T21:06:18+00:00

Narrative — prompt / model	describe-v1 / deepseek-chat	2026-08-18T20:41:28+00:00
Pipeline	0.1.0	2026-08-18

Generated 2026-08-19 04:58 UTC. Scores computed under a different weight set are not comparable with these (SC-10), and neither are market sizes computed under a different sizing version.